

AI-Based Digital Protection Relay for Transmission Line Fault Diagnosis Using CNN-BiLSTM

Abstract

The reliable operation of modern power transmission systems depends heavily on fast and accurate fault detection, classification, localization, and isolation. Conventional protection relays generally employ impedance-based calculations, overcurrent thresholds, and predefined operating characteristics to identify faults. However, these methods may experience reduced performance under nonlinear operating conditions, changing network configurations, and complex fault scenarios.

This project proposes an Artificial Intelligence-Based Digital Protection Relay using a hybrid Convolutional Neural Network (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) architecture for transmission line fault diagnosis in a 5-Bus power system. The proposed framework simultaneously performs three critical protection functions: fault type classification, fault location estimation, and relay trip decision generation.

The CNN module extracts spatial relationships among voltage and current measurements, while the BiLSTM module captures temporal dependencies associated with fault evolution. The learned feature representation is shared among three independent output heads responsible for fault classification, fault localization, and trip decision-making. The developed system represents an intelligent relay framework capable of supporting future smart-grid and digital protection applications.

Chapter 1: Introduction

Background

Transmission lines form the backbone of electrical power systems. Any disturbance

occurring on transmission lines can significantly affect system stability, reliability, and equipment safety.

Common transmission line faults include:

- Single Line-to-Ground (LG)
- Line-to-Line (LL)
- Double Line-to-Ground (LLG)
- Three-Phase Faults (LLL)
- Three-Phase-to-Ground Faults (LLLG)

When faults occur, abnormal current magnitudes and voltage distortions appear within milliseconds. If such faults remain undetected, severe consequences may occur including:

- Equipment damage
- Generator instability
- Transformer failures
- Cascading outages
- Large-scale blackouts

Therefore, fast and accurate fault diagnosis remains one of the most important functions in power system protection.

Need for Intelligent Digital Relays

Traditional relays rely on:

- Distance protection algorithms
- Differential protection schemes
- Overcurrent protection
- Impedance calculations

Although effective, these techniques require extensive setting calculations and may struggle under varying operating conditions.

Artificial Intelligence provides an alternative approach where the relay learns fault

characteristics directly from measured data rather than relying solely on predefined mathematical equations.

The integration of Deep Learning techniques allows the development of intelligent relays capable of:

- Automatic feature extraction
 - Real-time fault identification
 - Improved decision-making
 - Adaptive protection
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Chapter 2: Literature Review

Several researchers have explored machine learning and deep learning approaches for transmission line protection.

Previous studies have utilized:

- Artificial Neural Networks (ANN)
- Support Vector Machines (SVM)
- Long Short-Term Memory Networks (LSTM)
- CNN-LSTM Hybrid Models

The IEEE paper previously reviewed employed:

- ANN for fault classification
- LSTM for fault localization
- CNN-LSTM for enhanced fault location estimation

The study reported excellent fault diagnosis performance using machine learning techniques.

However, most existing works focus on only one or two protection functions.

Practical relay systems require:

1. Fault Identification
2. Fault Localization

3. Relay Tripping

within a single integrated framework.

This motivates the development of the proposed AI-Based Digital Protection Relay.

Chapter 3: Problem Statement

Conventional protection systems require multiple independent modules for:

- Fault Detection
- Fault Classification
- Fault Location Estimation
- Relay Operation

This increases system complexity and computational requirements.

The objective of this work is to develop a unified CNN-BiLSTM architecture capable of performing all major relay functions simultaneously using a shared feature-learning framework.

Chapter 4: Objectives

The primary objectives of the project are:

Objective 1

Develop a CNN-BiLSTM-based intelligent relay framework.

Objective 2

Accurately classify transmission line fault types.

Objective 3

Estimate fault location along the transmission line.

Objective 4

Generate automatic relay trip decisions.

Objective 5

Improve protection intelligence using deep learning.

Objective 6

Provide a scalable solution for future smart-grid protection systems.

Chapter 5: Dataset Description

Dataset Source

The dataset is stored in:

TLFaultDataset.xlsx

The dataset consists of fault and normal operating conditions collected from a transmission network.

Input Features

The input feature set contains electrical measurements such as:

- Phase Voltage Va
- Phase Voltage Vb
- Phase Voltage Vc
- Phase Current Ia
- Phase Current Ib
- Phase Current Ic

and other electrical parameters.

The following columns are removed from the feature matrix:

- Number
 - Type of Fault
 - Fault Location
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Output Targets

Target 1: Fault Type

Multi-class classification task.

Examples include:

- AG
 - BG
 - CG
 - AB
 - BC
 - AC
 - ABC
 - ABCG
 - No Fault
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Target 2: Fault Location

Continuous regression task.

Represents the distance of the fault from the relay location.

Target 3: Trip Decision

Binary classification task.

Trip Logic:

- 0 = No Trip
- 1 = Trip

This output directly emulates relay operation.

Chapter 6: Data Preprocessing

The preprocessing stage includes:

Dataset Inspection

- Data verification
- Shape analysis
- Data type checking

Missing Value Analysis

- Null value identification

- Data consistency verification

Label Encoding

Fault categories are transformed into numerical labels using LabelEncoder.

Train-Test Split

- Training Set = 90%
- Testing Set = 10%

Class Balancing

Sample weights are calculated using:

`compute_sample_weight()`

This prevents majority-class domination.

Feature Normalization

StandardScaler is applied to ensure:

- Faster convergence
- Stable optimization
- Improved model performance

Data Reshaping

Input format:

`(samples, time_steps, features)`

Current implementation:

`(samples, 1, features)`

Chapter 7: Proposed CNN-BiLSTM Digital Relay Architecture

The proposed architecture integrates CNN and BiLSTM layers within a multi-task learning framework.

CNN Feature Extraction Layer

Two convolutional blocks are used:

CNN Block 1

- Filters = 256
- Kernel Size = 1

CNN Block 2

- Filters = 256
- Kernel Size = 1

These layers learn fault-sensitive spatial representations from electrical measurements.

Feature Fusion Layer

Outputs from both CNN blocks are merged using concatenation.

Benefits:

- Rich feature extraction
 - Improved fault discrimination
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Bidirectional LSTM Layer

Configuration:

- Units = 256
- Return Sequences = True

The BiLSTM processes information in both directions:

- Forward Learning
- Backward Learning

This improves contextual understanding of fault behavior.

Shared Feature Representation

The extracted features are flattened and distributed to three independent prediction branches.

generalized fault-aware features, leading to improved predictive performance for both outputs.

Overall, the proposed CNN-BiLSTM-Attention framework demonstrates superior robustness, accuracy, and generalization capability, making it a promising candidate for

intelligent protection systems in future smart grid and microgrid applications.